On Performance of NNVC Inter-Coding

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***Abstract*—Over the past decades, video coding technologies have continuously advanced through increasingly sophisticated algorithms that push the limits of compression efficiency. However, further improvements using conventional techniques have become progressively more challenging. To address this, the ITU-T SG16 WP3 and ISO/IEC JTC 1/SC29 Joint Video Experts Team (JVET) established Ad-Hoc Group 11 (AHG11) to investigate neural network–based video coding (NNVC) methods. This paper elaborates on the development, implementation, and performance evaluation of the first JVET-adopted NNVC inter-coding tool based on Deep Reference Frame (DRF) model, including its combination tests with various NNVC tools. Specifically, the NNVC inter-coding tool DRF synthesizes a picture from two reconstructed frames in the Decoded Picture Buffer (DPB) and inserts it as the second reference frame in the reference picture list (RPL) to enhance motion estimation and compensation. Furthermore, the implementation of the DRF model on the Small Ad-hoc Deep-Learning Library (SADL) is also provided. In addition, combination tests with other NNVC tools are evaluated. Extensive experimental evaluations demonstrate the effectiveness of the NNVC inter-coding method. Compared with the NNVC-14.0 anchor under the Random Access (RA) configuration, the NNVC inter-coding tool achieves overall BD-rate gains of {2.20%, 1.46%, 2.22%} for the {Y, Cb, Cr} components, respectively.**

***Index Terms*—Inter prediction, reference frame generation, neural network, video coding**

I. Introduction

With the continuous evolution of video coding technologies, standards such as AVC/H.264 [1], HEVC/H.265 [2], and VVC/H.266 [3] have successively emerged, each improving coding efficiency through increasingly sophisticated algorithmic designs. However, the potential for further enhancement along this trajectory has become progressively limited. To address this challenge, the ITU-T SG16 WP3 and ISO/IEC JTC 1/SC29 Joint Video Experts Team (JVET) established Ad-Hoc Group 11 (AHG11) to investigate neural network–based approaches in video coding, a line of research known as NNVC [4]. Through successive iterations, tools such as neural network-based intra prediction (NN-Intra) [5] and neural network-based in-loop filter (NNLF) [5] have demonstrated substantial performance gains and have been incorporated into the NNVC reference software. Furthermore, the Small Ad-hoc Deep-Learning Library (SADL) [6], offering a straightforward framework that is dependency-free and compliant with the software development policy of JVET, has been adopted as the unified inference platform for all neural networks in NNVC.

This article elaborates the development, implementation, and performance evaluation of the first JVET-adopted NNVC inter-coding (NN-Inter) tool based on a Deep Reference Frame (DRF) model in response to the recent exploration efforts in JVET focusing on developing NNVC technologies beyond the capabilities of VVC. Specifically, the DRF core tool along with relevant NN-Inter supporting models are introduced, and exploratory studies that facilitated the development of NN-Inter tool are detailed. The DRF technique takes two reconstructed frames from the Decoded Picture Buffer (DPB) as inputs to synthesize a picture. The synthesized picture serves as an additional reference frame inserted into the reference picture list (RPL) at second place to provide a more reliable reference for subsequent motion estimation (ME) and motion compensation (MC). After several JVET meeting cycles of in-depth investigations, supported by multiple exploratory experiments, DRF technology has been integrated as an enhancement to conventional modules in the existing NNVC framework. In addition, to enable this advanced coding technology, the operator set of SADL was extended, which facilitated the int16 quantized deployment of the DRF model and provided technical support for broader neural network applications. Extensive experiments on top of the latest NNVC software (NNVC-14.0) have been conducted to evaluate the adopted NNVC inter-coding tool. Compared with NNVC-14.0, the NNVC inter-coding tool achieves overall {2.20%, 1.46%, 2.22%} BD-rate gains for {Y, Cb, Cr} under the Random Access (RA) configuration respectively.

II. History of NNVC inter coding development

The evolution of NNVC inter-coding reflects a continuous effort to leverage deep learning for enhanced video compression. As illustrated in Fig. 1, this field has progressed through several key developmental phases, culminating in the adoption of the Deep Reference Frame (DRF) model [9] within the NNVC software.

Early Explorations: Initial work in this domain, such as the DNN-based approach in [10], combined edge detection, optical flow estimation, and motion compensation modules, later refined with residual connections. Subsequently, NN-RFI methods adopted 3D-U-Net for virtual reference frame generation [11], exploring bidirectional frame interpolation within the VVC hierarchical structure and analyzing the impact of quantization parameters (QP) on frame quality—improving prediction accuracy without altering motion estimation or increasing bitstream overhead.

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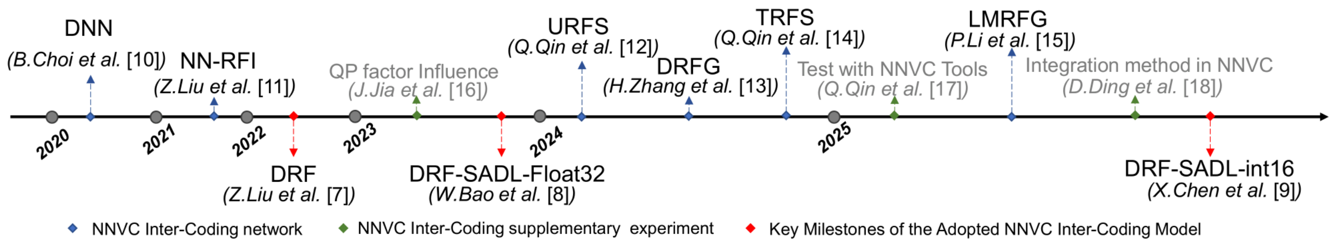

Fig. 1. Development timeline of the NNVC inter-coding tools, showing key milestones of the adopted DRF model (red) [7]–[9], initial proposals of related NNVC inter-coding models (blue) [10]–[15] and subsequent additional experiments (green) [16]–[18].

Development of DRF and Variants: The DRF framework, introduced in [7], extended earlier 3D-U-Net-based schemes from RA to Low-Delay B (LDB) configuration. Subsequent enhancements integrated advanced frame interpolation networks like IFRNet [19], [20] and simplified inputs by removing QP for better performance [16]. Engineering improvements were also critical: [8] implemented DRF using float32 SADL under RA configuration, while later works achieved inference acceleration through SIMD optimization [21], explored complexity-performance trade-offs [22], and ultimately realized the final int16 SADL implementation [9].

Additional Contributions to NNVC Inter-Coding Development: Several research lines provided valuable architectural alternatives for neural network-based inter coding. The Unified Reference Frame Synthesis (URFS) approach [12] introduced a unified learnable progressive-recursive motion estimation framework applicable to both RA and LDB configurations. Building upon this foundation, [17] further conducted preliminary exploration into the integration performance of URFS with other NNVC tools. The DRFG/LMRFG research line [13], [15] explored alternative network architectures that achieved competitive performance-complexity trade-offs. Meanwhile, the Transformer-based TRFS method [14] demonstrated the potential of attention mechanisms for reference frame synthesis tasks.

Final Adoption of DRF in NNVC: In pursuit of enhanced integration methodologies for NNVC inter-coding tools, [18] clarified that modifying the integration method by adding, rather than replacing, virtual reference frames in the codec's reference picture list yields better performance. Coupled with the implementation of int16 quantization for SADL-based DRF in [9], the DRF scheme demonstrated not only high efficiency but also ensured broad deployment compatibility across different platforms, leading to its final adoption in the NNVC software.

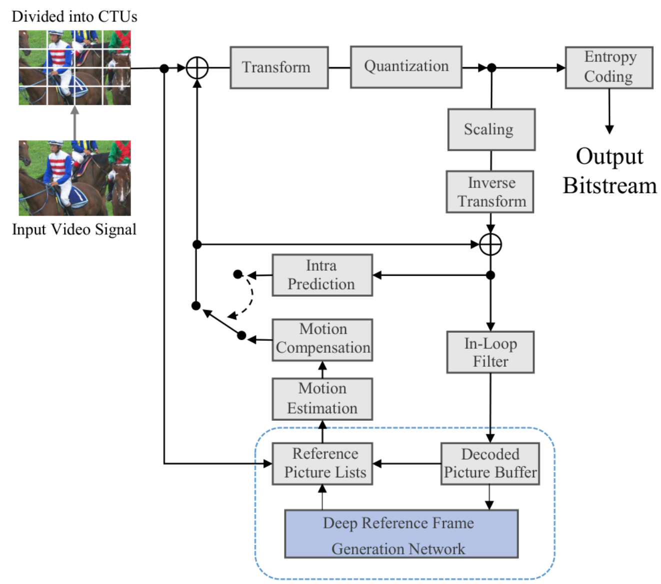

Fig. 2. The video encoding framework integrated with the Deep Reference Frame (DRF) generation method. The input frames derived from DPB are configured as two temporally equidistant frames for bi-directional prediction. A high-quality, synthesized reference frame is generated by DRF model and integrated into the reference picture lists to enhance inter prediction.

III. The DRF and SADL Implementation

A. Deep Reference Frame Generation

As illustrated in Fig. 2, DRF extracts frames from the codec's DPB as input, generating a high-quality deep reference frame to provide a high-quality reference frame for the current encoding frame.

Due to the hierarchical coding structure of the codec under the RA configuration, and considering that frames in closer temporal distance exhibit stronger temporal correlation, the DRF method is restricted to frames at temporal layers 3, 4, and 5. When applied, the relevant frames used are positioned at distances of 4 frames, 2 frames, and 1 frame from the target encoded frame, respectively. The formula is shown below:

$$\begin{aligned} ([\mathrm{F}]\mathrm{YUV}, [\mathrm{B}]\mathrm{YUV}) &= \left(I^r_{p-\delta\mathrm{p}}, I^r_{p+\delta\mathrm{p}}\right), \\ \text{where } \delta p &= \begin{cases} 4, & L\left(I_p\right)=3, \\ 2, & L\left(I_p\right)=4, \\ 1, & L\left(I_p\right)=5, \end{cases} \end{aligned} \tag{1}$$

where $([\mathrm{F}]\mathrm{YUV}, [\mathrm{B}]\mathrm{YUV})$ correspond to the preceding (Forward, F) and subsequent (Backward, B) reference frames in display order. I_p is the current frame identified by its Picture Order Count (POC) p, and I^r_p is the corresponding reconstructed frame. The notation $L(\cdot)$ signifies the temporal layer of a frame. Furthermore, δp refers to the smallest available POC distance separating the current frame I_p from the encoded frames adjacent to it in the sequence.

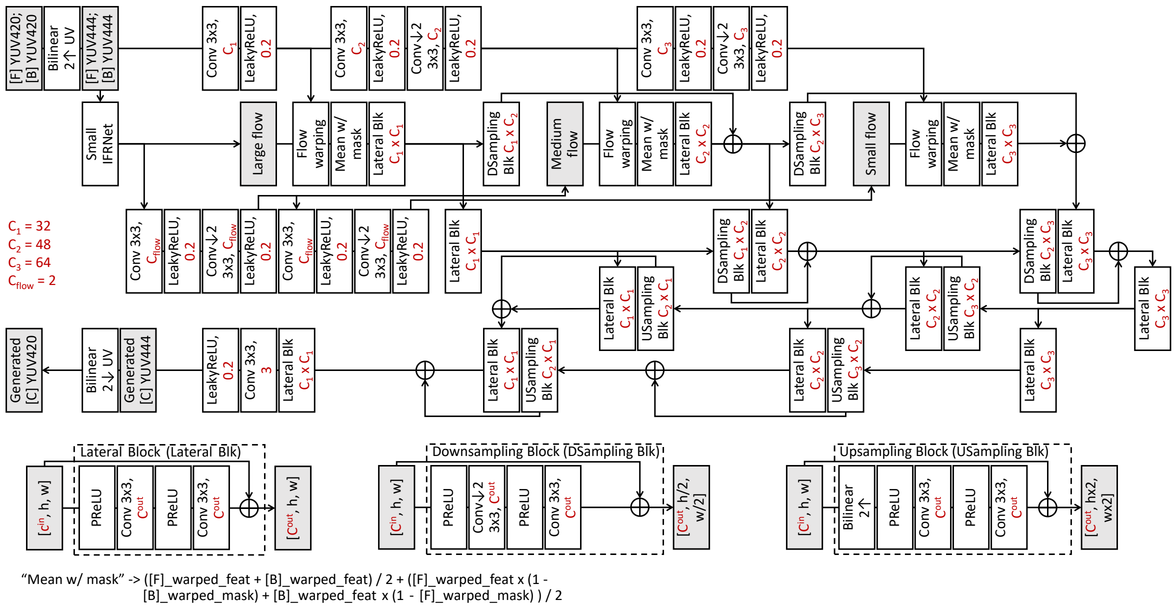

Fig. 3. The architecture of the deep reference frame generation network [9].

Since the deep reference frame shares the same POC as the target encoding frame, the temporal distance between the current frame and its deep reference frame becomes zero. This configuration invalidates conventional Temporal Motion Vector Prediction (TMVP) techniques and their derivatives, which rely on non-zero temporal distances for motion vector scaling. To ensure coding stability and maintain bitstream consistency, the codec explicitly disables TMVP-related mechanisms when the reference frame of a coding block or its collocated block is a deep reference frame during inter prediction.

The architecture of the deep reference frame generation model is illustrated in Fig. 3. Reconstructed YUV420 frames from the DPB are first upsampled to YUV444 via bilinear interpolation. After initial upsampling for YUV resolution alignment, multi-scale features are extracted via a three-level convolutional module. YUV444 frames are converted to RGB for accurate bidirectional optical flow estimation. Bidirectional optical flows are estimated by IFRNet [19] and fused using a masked blending operation. The forward and backward flows are used to warp YUV-domain features, which are adaptively fused and refined through a GridNet [23] enhancement module for final frame synthesis. Despite the compression artifacts in the reconstructed inputs, the DRF network effectively performs temporal interpolation while enhancing the quality of the generated reference frame.

B. SADL Implementation of DRF

To enable both float32 and int16 inference for the DRF model in SADL, multiple new operators were introduced and existing operators were optimized as extensions to the SADL library [24], [25]. The conversion function from PyTorch to SADL was modified accordingly. Layer-wise quantization parameters for the DRF model were determined using the quantization tools provided by the SADL library. Through these enhancements, efficient deployment of the int16 DRF model was achieved [9]. The main updates and new features of SADL are summarized in Table I.

TABLE I
SUMMARY OF SADL CHARACTERISTICS WITH NEWLY INTRODUCED FEATURES EMPHASIZED

Language	Pure C++, header-only
Optimization	SIMD at hotspots
Compatibility	ONNX to SADL converter
New features	Gridsample, ScatterND, Resize, Where, Compare
Type support	float, int32, int16
License	BSD 3-Clause

IV. EXPERIMENTS

To comprehensively evaluate the performance of the NNVC inter-coding tool (DRF), experiments were conducted using the latest NNVC reference software (NNVC-14.0). The evaluation was carried out on JVET common test conditions [4] to assess both coding efficiency and computational complexity. The overall impact of the DRF method was further analyzed under the NNVC-14.0 default configuration, where the NN-Intra module and the NN in-loop filter (NNLF) operating at low-complexity operation point (LOP6), were enabled. An ablation study was subsequently conducted to evaluate the individual and combined contributions of the NN-Inter tool DRF, NN-Intra, and LOP6 in terms of performance and complexity.

TABLE II
OVERALL PERFORMANCE OF THE DRF IN NNVC-14.0 OVER NNVC-14.0 (WITH NN-INTRA AND LOP6 ENABLED)

Class	BD-Rate			Complexity	
	Y	U	V	EncT	DecT
A1	-1.41%	-0.61%	-2.01%	160%	2483%
A2	-2.50%	-1.14%	-2.26%	148%	2284%
B	-1.81%	-1.33%	-1.62%	155%	2643%
C	-3.07%	-2.50%	-3.09%	136%	2412%
Overall	-2.20%	-1.46%	-2.22%	149%	2474%
D	-3.21%	-0.50%	-1.67%	137%	2338%
F	-0.99%	-0.65%	-0.35%	181%	5968%

TABLE III
PERFORMANCE OF VARIOUS NNVC TOOL COMBINATIONS COMPARED TO NNVC-14.0 (WITH NNVC TOOLS DISABLED)

NN-Inter		✓			✓	✓		✓
NN-Intra			✓		✓		✓	✓
LOP6				✓		✓	✓	✓
ClassB	Y	-1.88%	-1.87%	-6.17%	-3.76%	-7.63%	-7.73%	-9.19%
	U	-2.04%	-1.28%	-14.55%	-3.39%	-15.30%	-15.97%	-16.71%
	V	-1.83%	-1.55%	-13.30%	-3.39%	-13.44%	-14.72%	-14.82%
	EncT	165%	112%	106%	181%	173%	116%	183%
	DecT	59119%	138%	2298%	60124%	62212%	2219%	60346%
ClassC	Y	-3.70%	-1.72%	-6.05%	-5.39%	-8.83%	-7.47%	-10.23%
	U	-3.29%	-1.52%	-13.03%	-4.88%	-14.62%	-14.37%	-15.97%
	V	-3.33%	-1.86%	-12.17%	-5.30%	-13.63%	-13.74%	-15.21%
	EncT	141%	114%	103%	157%	147%	114%	157%
	DecT	52206%	160%	2206%	54308%	56350%	2231%	54738%
ClassD	Y	-4.47%	-1.38%	-6.83%	-5.84%	-9.64%	-8.04%	-10.82%
	U	-2.39%	-1.24%	-12.43%	-3.73%	-12.21%	-13.75%	-13.57%
	V	-2.33%	-1.72%	-11.15%	-3.85%	-10.72%	-12.58%	-12.17%
	EncT	136%	112%	100%	153%	141%	106%	144%
	DecT	43871%	166%	2003%	45622%	47534%	1897%	42591%
Average	Y	-3.35%	-1.66%	-6.35%	-4.99%	-8.70%	-7.75%	-10.08%
	U	-2.57%	-1.35%	-13.34%	-4.00%	-14.04%	-14.70%	-15.42%
	V	-2.50%	-1.71%	-12.20%	-4.18%	-12.60%	-13.68%	-14.07%
	EncT	147%	113%	103%	164%	154%	112%	161%
	DecT	51732%	155%	2169%	53351%	55365%	2116%	52558%

A. *Overall Results*

Table II summarizes the overall performance of the NNVC inter-coding method compared with the NNVC-14.0 default configuration, where NN-Intra and the LOP6 are enabled. It can be observed that the NNVC inter-coding tool consistently achieves notable BD-rate gains across all test classes, confirming its effectiveness in improving coding efficiency. Although the inclusion of neural network inference inevitably increases both encoding and decoding complexity, the achieved gains demonstrate a favorable balance between compression efficiency and computational cost within the NNVC framework.

B. *Ablation Tests*

Table III also summarizes the BD-rate performance achieved by various combinations of the neural network-based coding tools. The following observations can be made from Table III:

- **NN-Inter (DRF) and NN-Intra:** The combination of NN-Inter and NN-Intra produces nearly additive BD-rate improvements. This behavior can be attributed to their independent and parallel operation, as they target different stages of the coding process without functional overlap.
- **NN-Inter and LOP6:** In contrast, a noticeable performance overlap is observed between LOP6 and NN-Inter. This is because LOP6 enhances both the target and reference frames through filtering, while NN-Inter synthesizes virtual reference frames to improve reference quality—both aiming at reference enhancement. Moreover, since NN-Inter operates on frames processed by LOP6, a dependency arises between them, thereby limiting the incremental gains when they are combined.
- **NN-Intra and LOP6:** The coding gains of LOP6 and NN-Intra also exhibit near-additive behavior, indicating a complementary relationship. This is primarily attributed to their distinct operational domains: NN-Intra specifically optimizes intra-coded frames and blocks, while LOP6 enhances reconstructed quality across various frame types through its filtering mechanism. This functional separation ensures minimal interference between the two tools, allowing their individual contributions to accumulate effectively in the overall coding performance.

Besides the coding performance, Table III also presents the encoding and decoding time results for various combinations of the NN-based coding tools. The following observations can be made from Table III:

- **Encoding Complexity:** Among the three tools, NN-Inter exhibits the highest encoding time, followed by LOP6, while NN-Intra demonstrates the lowest. The encoding times of these tools show a largely additive trend. Although NN-based tools may slightly influence the selection behavior of rate–distortion optimization (RDO)-related modules within the codec, the primary factor contributing to the time increase lies in the intrinsic computational cost of network inference. Overall, the encoding complexity demonstrates an approximately additive effect across different tool combinations.
- **Decoding Complexity:** In terms of decoding time, NN-Inter again exhibits the highest complexity, followed by LOP6, while NN-Intra remains the most lightweight. A slight negative correlation is observed between the decoding times of NN-Intra and LOP6, mainly because both tools enhance the reconstructed frame quality, thereby reducing the data workload of the decoder. In contrast, the impact of NN-Inter is primarily additive, as its computational overhead significantly outweighs its influence on the codec pipeline.

V. CONCLUSIONS

This paper presented the development of the first NNVC inter-coding tool adopted by JVET, introducing the Deep Reference Frame (DRF) model and the extended SADL operator library. The joint performance of neural network-based coding tools, including NN-Inter (DRF), NN-Intra, and NNLF (LOP6), was also evaluated. Experimental results effectively validate the superiority of the NNVC inter-coding techniques, demonstrating overall BD-rate reductions of {2.20%, 1.46%, 2.22%} for the {Y, Cb, Cr} components, respectively, under the RA configuration, demonstrating the DRF's strong potential for future NNVC development.

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